

Paper III — Certified-Constrained MARL

Certificates as Constraints: Feasibility-Preserving Multi-Agent Reinforcement Learning for Swarms Whose Guarantees Degrade

Research proposal from the CERTIFLIGHT concept (WP4) · RobustUAVs.ai · Roger Nick Anaedevha · 2026-10-01
Target: NeurIPS, main track · builds on the parent manuscript [5] and Papers I–II

Abstract

Safe reinforcement learning offers two kinds of guarantee and neither is the one a safety-critical fleet needs. Lagrangian methods satisfy constraints *in expectation* over a horizon; barrier and shielding methods give *pointwise* guarantees but read their constraint sets from hand-specified margins. We study the case where the constraint set is instead the output of a formal robustness certificate that is itself *time-varying and observation-dependent*: each agent carries a certified navigation radius $R_i(t)$ that shrinks as the agent is jammed or faulted, and may hold a task only while $R_i(t)$ exceeds the radius that task provably requires. This couples a certificate to a constrained Markov decision process in a way the safe-RL literature has not analysed, because the feasible set moves as a function of the observation the policy is still learning to act on. We give three results: a projection safety layer that preserves pointwise feasibility at every step of *both learning and deployment* whenever a feasible assignment exists; an exact, checkable condition for when one exists, via a deficiency form of Hall’s theorem; and a bound on coverage loss as a function of the fraction of agents whose certificate has collapsed, matching a brute-force oracle in simulation. The contribution is not a new RL algorithm but a correctness analysis of certificate-as-constraint, with the certificate taken as an adversarially moving object rather than a fixed box.

1 The problem the existing guarantees miss

A swarm flies a coverage mission — urban-corridor monitoring, environmental transects, point-to-point medical delivery. Each aircraft’s navigation stack carries a certificate (the parent manuscript’s [5], or any producing a radius $R_i(t)$: the largest position error under which the aircraft still completes its assigned task). Under jamming or a sensor fault, $R_i(t)$ shrinks. Below the radius a task requires, the aircraft can no longer be trusted with it, and the fleet must re-allocate.

The decision — which aircraft holds which task, as certificates move — is a constrained sequential control problem. The safe-RL literature has the pieces and not the composition:

- **Lagrangian constrained RL** [1, 9] satisfies constraints *in expectation over the horizon*. A fleet that violates a safety radius on 5% of steps but satisfies it on average is not acceptable; expectation is the wrong quantifier for flight.
- **Shielding and control barrier functions** [2, 4] give pointwise guarantees, but from a *hand-specified* safe set. The margin is an engineering choice disconnected from any formal statement about the perception stack.
- **Certified RL** [11, 12] certifies a policy against an observation-perturbation ball, single-agent, with the ball fixed in advance. It never feeds a certificate into a *fleet-level* constraint.

The gap is specific: **no analysis treats the constraint set of a multi-agent policy as the output of a formal certificate that varies with the observation**. That is the object here, and it is harder than a fixed constraint because the feasible region the policy must stay inside is itself a function of the jamming the policy is reacting to.

2 Setup

A fleet of N agents and M tasks is a constrained Markov decision process [3] $(\mathcal{S}, \mathcal{A}, P, r, \{c_i\})$. Agent i ’s state includes its certified radius $R_i(t)$, computed online by the perception certificate from locally

estimated jamming and fault conditions. An allocation π assigns agents to tasks to maximise coverage, subject to the pointwise constraints

$$c_i : R_i(t) \geq R_{\min}(\tau_i) \quad \text{whenever agent } i \text{ holds task } \tau_i, \text{ at every } t. \quad (1)$$

We take $R_i(t)$ as given by the certificate and treat it, adversarially, as an arbitrary bounded cadlag process — no distributional assumption, because a jammer is not a distribution. $R_{\min}(\tau)$ is the task’s required radius, from the mission geometry.

3 Three results

3.1 Feasibility preservation (T3)

Proposition 1 (Projection preserves pointwise feasibility). *Let $\mathcal{F}(t)$ be the set of assignments satisfying (1) at the current radii $R_i(t)$. If the policy’s proposed assignment a_t is replaced by its projection $\Pi_{\mathcal{F}(t)}(a_t)$ onto the feasible set whenever $\mathcal{F}(t) \neq \emptyset$, then every executed assignment satisfies (1) pointwise, during learning and deployment alike, regardless of the policy’s parameters.*

The content is not the projection — that is immediate once $\mathcal{F}(t)$ is known — but that it holds *during learning*, so a half-trained policy never flies an infeasible assignment, and that $\mathcal{F}(t)$ is cheap to represent (next section). This is what lets a certificate replace a hand-tuned barrier margin: the margin becomes a computed, auditable quantity.

3.2 When a feasible assignment exists (T3, exactly)

Feasibility is a bipartite matching: an edge joins agent i to task j iff $R_i(t) \geq R_{\min}(\tau_j)$. A full assignment exists iff the matching saturates the tasks, and Hall’s theorem [7] gives the condition. We use its deficiency form, specialised to the *nested* structure that (1) forces — an agent that qualifies for a demanding task qualifies for every easier one — which collapses the exponential Hall check to a linear one.

Proposition 2 (Checkable feasibility). *Sort task requirements descending, $\rho_1 \geq \dots \geq \rho_M$, and let $n(\rho) = |\{i : R_i(t) \geq \rho\}|$. The maximum number of tasks simultaneously coverable is $M - d$, where the deficiency is*

$$d = \max_{1 \leq k \leq M} (k - n(\rho_k))^+.$$

A full assignment exists iff $d = 0$, checkable in $O(N + M \log M)$.

Proposition 3 (k -robust coverage). *Full coverage survives the collapse of any k agents’ certificates iff $n(\rho_j) \geq j + k$ for every j . The largest such k is the fleet’s coverage margin: the number of simultaneous certificate failures it tolerates with no coverage loss.*

■ *Both propositions are elementary once the nested structure is seen; the contribution is seeing it, and that it makes an otherwise NP-flavoured fleet-robustness question a linear-time check an onboard planner can run every control step. We verified both against a brute-force matching oracle over 3000 random instances: exact agreement on the coverage count and on the k -robust condition. That is a sanity check on the math, not evidence about flight.*

3.3 Coverage loss as certificates degrade (T4)

Proposition 4 (Coverage-degradation bound). *If a fraction ϕ of agents have $R_i(t)$ below the median task requirement, coverage loss is at most $\lceil \phi N \rceil$ tasks, and exactly $M - (M - d_\phi)$ with d_ϕ the deficiency of Proposition 2 recomputed on the degraded radii. The resulting curve, coverage versus ϕ , is the operational trade-off between per-agent robustness investment and the fleet size needed for a target coverage guarantee.*

Classical coverage control has results for *failed* (binary) agents [6]; the novelty is a bound parameterised by *continuous* certified-capability degradation, which is what a procurement decision turns on — how many aircraft, carrying how good a navigation stack, to hold a coverage SLA under a given jamming level.

4 What is proposed, and what is already in hand

In hand	To build
A certificate producing $R_i(t)$ from jamming/fault state (parent manuscript; $\gamma = 1.625$ m/s campaign supremum)	the CMDP, the projection layer, and the learned allocation policy
A physics-informed fleet simulator, MAX_FLEET=12, ten attack classes	a jamming/fault/gust schedule driving $R_i(t)$ per agent over the mission
Propositions 1–4, brute-force-checked	the empirical study: feasibility violations, coverage curves, scaling

5 Experimental plan

- RQ1** Does the projection layer achieve *zero* pointwise violations during training, where a PPO-Lagrangian baseline [13, 1] achieves only in-expectation satisfaction? Report per-step violation counts, not episode averages.
- RQ2** How does the measured coverage-versus- ϕ curve compare to the bound of Proposition 4, across jamming intensities and fault and gust profiles?
- RQ3** How does the certified-constraint policy compare to jamming-blind and worst-case-fixed-margin baselines on coverage and on completion?
- RQ4** At what fleet size and task count does the projection step stop being real-time? Report the scalability ceiling rather than hiding it.

Baselines. Jamming-blind MAPPO [13]; PPO-Lagrangian and CPO [1]; shielded MADDPG [8, 2]; QMIX [10]; and a worst-case fixed-margin allocator. The point of the comparison is to separate what the *learned policy* achieves from what the *guarantee mechanism* enforces — so every method is reported both with and without the projection layer.

Why NeurIPS and not a robotics venue. The result is about the quantifier on a learned constraint — pointwise versus in-expectation, under a constraint set that is itself a certified, adversarially moving object. That is a learning-theory and safe-RL contribution. The UAV swarm is the setting that makes it concrete and gives it a benchmark, not the contribution itself.

6 Relation to the other papers and to the parent

- **Parent manuscript / Paper I** supply $R_i(t)$. Paper III consumes it. The interface is exactly the certified radius; nothing about how it is computed leaks into the CMDP analysis, which is why Paper III stands if the certificate is swapped.
- **Paper II (SoK)** places this work at its $\mathbf{Q}_{op}/\mathbf{Q}_{pop}$ boundary: pointwise feasibility is a per-operation guarantee, the coverage bound a per-population one. Paper III is a worked instance of a sound $\mathbf{Q}_{op} \rightarrow \mathbf{Q}_{pop}$ lift.
- **Distinct from all three** in object: Papers I, II and the parent are about a certificate; Paper III is about a *controller that consumes* one.

7 Risks

- **The analysis could outrun the experiments.** The three propositions are the contribution and are already proved and checked; the RL study is supporting evidence. If MARL training is unstable at the target fleet size, the paper reports the projection layer’s guarantees (which hold for *any* policy, including a random one) and the scalability ceiling, and still stands.

- $R_i(t)$ is only as good as the certificate. Paper III assumes it; if the certificate is loose, constraints are conservative and coverage suffers, but feasibility and the bounds are unaffected. This dependence is stated, not hidden.
- **Sim-to-real.** The study is in simulation. The host’s UAV testbed is the path to a hardware vignette; it is a strengthener, not a dependency, and over-claiming transfer from simulation is the failure mode to avoid.
- **Novelty challenge (“this is just matching + shielding”).** Pre-empt it: the novelty is the *coupling* — a formal perception certificate as a time-varying CMDP constraint with feasibility preserved *during learning* — and the linear-time feasibility and k -robustness conditions that the nested structure of certified radii makes possible.

8 Page plan and timeline

Section	pp	Section	pp
Introduction	1.0	Experiments	2.5
Related work	1.0	Discussion and limitations	0.75
Setup (CMDP, certificate)	1.0	Conclusion	0.25
Theory (Prop. 1–4)	2.0	(appendix: proofs, uncapped)	—
Body, within a 9-page NeurIPS limit			8.5

About four to five months, and the theory is done, so writing can begin while the RL study runs. The simulator and certificate exist; the new engineering is the CMDP wrapper, the projection layer and the degradation schedule.

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